

Explainable Financial Alerts for Retail Investors: Integrating Statistical Anomaly Detection and News–Market Impact Correlation

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**A dissertation submitted in partial fulfillment of
the requirements for the degree of Master of Science,
Specialisation Area of**

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Porto, June 21, 2026

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Dedictory

Abstract

This dissertation presents an explainable (XAI-first) financial-alert system for retail investors in the US equity market (NYSE and NASDAQ). The system raises Telegram alerts from two independent triggers — abrupt market movements, detected as statistical anomalies, and incoming financial news — and, for every alert, exposes a fully traceable explanation: the detected event, the reasoning applied, the sources, and historical precedents. Its core is a news–market correlation engine that, given a news item, retrieves analogous historical news from a knowledge base built on the FNSPID dataset and measures the impact those precedents had on prices. As an Artificial Intelligence Engineering contribution, the work integrates, applies and critically evaluates existing components in a functional, explainable and reproducible system, with a documented news-impact correlation methodology.

Keywords: Explainable AI, Financial Alerts, Anomaly Detection, News Impact, Retail Investors, Financial NLP

Resumo

Esta dissertação apresenta um sistema de alertas financeiros explicável (XAI-first) para investidores de retalho no mercado acionista dos Estados Unidos (NYSE e NASDAQ). O sistema envia alertas via Telegram a partir de dois gatilhos independentes — movimentos abruptos de mercado, detetados como anomalias estatísticas, e novas notícias financeiras — e, para cada alerta, expõe uma explicação totalmente rastreável: o evento detetado, o raciocínio aplicado, as fontes e os precedentes históricos. O seu núcleo é um motor de correlação notícia–mercado que, dada uma notícia, recupera notícias históricas análogas de uma base de conhecimento construída sobre o conjunto de dados FNSPID e mede o impacto que esses precedentes tiveram nos preços. Enquanto contribuição de Engenharia de Inteligência Artificial, o trabalho integra, aplica e avalia criticamente componentes existentes num sistema funcional, explicável e reproduzível, com uma metodologia documentada de correlação notícia–impacto.

Palavras-chave: Explainable AI, Financial Alerts, Anomaly Detection, News Impact, Retail Investors, Financial NLP

Acknowledgement

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List of Algorithms

List of Source Code

List of Symbols

r_t	log-return of an asset on day t	—
μ_t	rolling mean of returns	—
σ_t	rolling standard deviation of returns	—
z_t	z-score of the return on day t	—
k	anomaly threshold (in standard deviations)	—
w	rolling-window length	d

List of Acronyms

AI	Artificial Intelligence.
NASDAQ	National Association of Securities Dealers Automated Quotations.
NYSE	New York Stock Exchange.
SIFMA	Securities Industry and Financial Markets Association.
XAI	Explainable Artificial Intelligence.

Chapter 1

Introduction

This chapter introduces the dissertation. It motivates the work, states the problem, defines the objectives and research questions, summarises the scientific contributions, and outlines the structure of the document.

1.1 Motivation and Overview

The United States equity market is the largest in the world and is owned, directly or indirectly, by a majority of US households: 62% of Americans reported owning stock in 2025 (Gallup 2025), in a market whose capitalisation reached US\$62.2 trillion at the end of 2024 (Securities Industry and Financial Markets Association (SIFMA) 2025). At the same time, artificial intelligence has been adopted rapidly across financial services (Cambridge Centre for Alternative Finance 2026), yet many of the resulting models behave as opaque “black boxes” whose reasoning users cannot inspect (Barredo Arrieta et al. 2020). Chapter 2 develops this context in detail.

For the non-professional (“retail”) investor, this combination is difficult. Markets move continuously across thousands of tickers and financial news arrives without pause, but the tools that help to interpret these signals are typically institutional, opaque, or both. This dissertation addresses that gap with an *explainable* financial-alert system: one that not only flags relevant events but also explains them in a way the investor can follow, verify and act upon.

1.2 Problem Statement

A retail investor who wishes to stay informed faces two distinct problems. First, distinguishing a genuinely abnormal price movement from ordinary volatility requires statistical judgement and constant attention. Second, when a relevant news item appears, assessing its likely impact requires knowledge of how analogous past events affected prices — information that is rarely at hand. Existing consumer tools tend to deliver alerts without explanation, or rely on models whose internal logic is hidden, which is precisely the transparency problem that explainable AI seeks to address (Adadi and Berrada 2018; Barredo Arrieta et al. 2020).

The problem this work tackles is therefore not price prediction, but *explanation*: detecting abrupt market movements and relevant news, and presenting, for each alert, a traceable account of what was detected, the reasoning applied, the sources, and concrete historical precedents of similar events and their observed impact.

1.3 Objectives and Research Questions

The general objective is to design, implement and evaluate an explainable, reproducible financial-alert system for US retail investors, driven by two independent triggers (abrupt market movements and incoming news) and delivered through Telegram. This is pursued through three research questions:

- **RQ1.** Can transparent statistical methods detect abrupt market movements with enough precision to be useful to a retail investor, while remaining explainable?
- **RQ2.** Can a news–market correlation engine retrieve genuinely analogous historical news and quantify their observed price impact, without lookahead, better than trivial baselines?
- **RQ3.** Can the resulting explanations be made faithful to the system’s actual logic and useful to a retail investor?

The supporting objectives are: (i) build a thin end-to-end slice early; (ii) develop each component in its simplest defensible form; (iii) evaluate each component against a design defined beforehand (Chapter 6); and (iv) keep the whole system reproducible and transparent.

1.4 Scientific Contributions

This is an Artificial Intelligence *Engineering* dissertation. The contribution is not to invent new algorithms but to *integrate, apply and critically evaluate* existing components in a functional, explainable and reproducible system. Concretely, the contributions are:

- A documented, reproducible methodology for **news–market impact correlation**, combining sentence embeddings for retrieval with event-study windows for impact measurement, with explicit choices of similarity metric and window and explicit avoidance of lookahead.
- An **XAI-first alerting pipeline** that exposes its full reasoning chain (detection, evidence, sources, precedents) to a non-expert user.
- A **critical evaluation** of each component against baselines and a transparent design, reporting honest results and limitations.

1.5 Document Structure

The remainder of the document is organised as follows. Chapter 2 sets the market and technological context. Chapter 3 reviews the relevant literature on anomaly detection, explainable AI and financial NLP, with comparative tables. Chapter 4 describes the system architecture and the method chosen for each component. Chapter 5 details the implementation. Chapter 6 reports the evaluation. Chapter 7 concludes with the main findings, limitations and future work.

Chapter 2

Contextualization

This chapter sets the context for the work. It outlines the scale of the United States equity market, the growth of retail investing, the rapid adoption of Artificial Intelligence (AI) in finance and the explainability concerns it raises, and the information-overload problem that motivates an explainable financial-alert system.

2.1 The US Equity Market (NYSE and NASDAQ)

The United States equity market is the largest and among the deepest and most liquid in the world. According to the Securities Industry and Financial Markets Association (SIFMA) Capital Markets Fact Book, US equity markets represented 49.1% of the US\$126.7 trillion in global equity market capitalisation at the end of 2024 — about US\$62.2 trillion, some 5.3 times the size of the next-largest market, China (Securities Industry and Financial Markets Association (SIFMA) 2025). The two principal trading venues are the New York Stock Exchange (NYSE) and National Association of Securities Dealers Automated Quotations (NASDAQ), which together list the large-capitalisation companies that dominate global indices.

This market has also grown substantially over the last decade. As shown in Figure 2.1, US equity market capitalisation rose from US\$25.1 trillion in 2015 to US\$62.2 trillion in 2024 (Securities Industry and Financial Markets Association (SIFMA) 2025), a scale and liquidity that make it the natural focus for a retail-oriented alerting system.

2.2 The Retail-Investing Landscape

Equity ownership is widespread among US households. In 2025, 62% of Americans reported owning stock — whether individual shares or holdings within mutual funds and retirement accounts — matching the 2024 figure and continuing a rebound after more than a decade below that level (Gallup 2025). Ownership is, however, unevenly distributed: it rises to 87% among households earning US\$100,000 or more and falls to 28% among those earning under US\$50,000 (Gallup 2025).

Beyond ownership, individual (“retail”) investors have become a material part of day-to-day trading activity, helped by commission-free mobile brokerages. Published estimates of the retail share of US equity trading volume vary widely with methodology and period — broadly in the range of 20–35% during 2024–2025. This combination of broad ownership and growing direct participation defines the audience this work serves: the non-professional investor who must make sense of the market without the tools available to institutions.

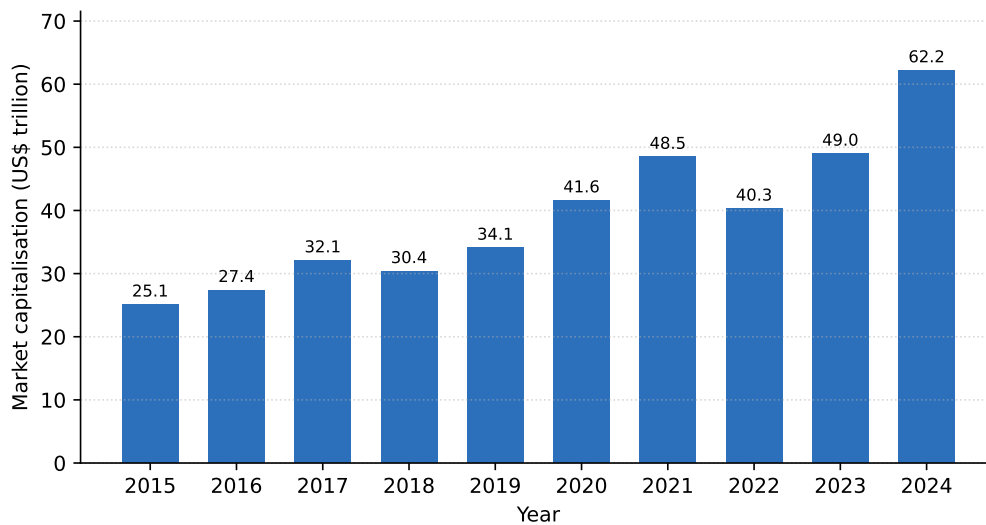


Figure 2.1: US equity market capitalisation, 2015–2024 (US\$ trillion). Figure generated from data reported in the SIFMA 2025 Capital Markets Fact Book (Securities Industry and Financial Markets Association (SIFMA) 2025).

2.3 Artificial Intelligence in Finance

AI has been adopted rapidly across financial services. A 2026 industry survey by the Cambridge Centre for Alternative Finance found that 81% of surveyed financial-services firms were adopting AI at some level, 40% of industry respondents reported advanced adoption, and 71% were already using generative AI (Cambridge Centre for Alternative Finance 2026). Typical applications include fraud detection, market analysis and customer service.

This rapid adoption raises a well-documented concern: many high-performing models behave as “black boxes”, offering little insight into how a given output was produced. Explainable Artificial Intelligence (XAI) addresses precisely this gap, seeking methods whose reasoning users can follow and trust (Adadi and Berrada 2018; Barredo Arrieta et al. 2020). For a retail investor, who must act on — and bear the consequences of — a system’s output, this transparency is not a luxury but a requirement. It is the principle that guides the present work.

2.4 The Information-Overload Problem

A retail investor faces two simultaneous streams of information: continuous price movements across thousands of US tickers, and a constant flow of financial news. Distinguishing a meaningful movement from ordinary volatility, and understanding whether and how a news item is likely to affect an asset, requires time, data and expertise that most non-professionals do not have.

This is the gap the dissertation addresses. Rather than predicting prices, the proposed system detects abrupt movements and relevant news, and — crucially — explains them: what was detected, the reasoning applied, the sources, and concrete historical precedents of analogous events and their observed impact. The remainder of the document develops and evaluates this explainable alerting system.

Chapter 3

Literature Review

This chapter reviews the literature underpinning the system. It covers anomaly detection in financial markets, explainable artificial intelligence, financial natural language processing, and event-study approaches to news–market impact. Each area combines seminal and recent work, and comparative tables summarise the approaches, their mechanisms and their limitations. The chapter closes by positioning the present work and identifying the gap it addresses.

3.1 Anomaly Detection in Financial Markets

Anomaly detection is a mature field with techniques spanning many domains. The survey of Chandola, Banerjee, and Kumar (2009) provides the standard taxonomy — point, contextual and collective anomalies — and groups methods into statistical, distance-based, and machine-learning families. For financial returns, simple statistical methods (deviations from a rolling mean and standard deviation, i.e. a z-score) are attractive because they are transparent and easy to justify, which is an advantage for an explainable system. More expressive machine-learning detectors exist, such as Isolation Forest (Liu, Ting, and Zhou 2008), which isolates anomalies through random partitioning and scales well to large, high-dimensional data; the trade-off is reduced direct interpretability. Table 3.1 compares the two families.

Table 3.1: Anomaly-detection approaches relevant to this work.

Approach	How it works	Strengths / limitations
Statistical (rolling) (Chandola, Banerjee, and Kumar 2009)	z-score Flags returns that deviate from a rolling mean/standard deviation	Transparent and easy to explain (XAI); assumes locally stable distribution
Isolation Forest (Liu, Ting, and Zhou 2008)	Isolates points via random partitioning (ensemble)	Scalable, handles high dimensionality; less directly interpretable

3.2 Explainable Artificial Intelligence (XAI)

As AI systems are deployed in high-stakes settings, the inability to inspect their reasoning has become a central concern. The surveys of Adadi and Berrada (2018) and Barredo Arrieta et al. (2020) map the field: its motivations, a taxonomy of transparent versus post-hoc methods, and the open challenges toward responsible AI. Among post-hoc, model-agnostic

techniques, two are widely used. LIME (Ribeiro, Singh, and Guestrin 2016) explains a single prediction by fitting a simple surrogate model in its local neighbourhood; it is intuitive but its explanations can be unstable and are only locally faithful. SHAP (Lundberg and Lee 2017) attributes a prediction to its input features using Shapley values, offering consistency and a theoretical grounding at a higher computational cost. Table 3.2 summarises these. In this work, explainability is pursued primarily through *transparent design* (statistical detection and retrieved precedents), with feature attribution as an optional complement.

Table 3.2: Explainability approaches considered.

Method	Type	Strengths / limitations
LIME (Ribeiro, Singh, and Guestrin 2016)	Local surrogate, model-agnostic	Intuitive; explanations can be unstable, only locally faithful
SHAP (Lundberg and Lee 2017)	Shapley-value attribution, model-agnostic	Consistent, theoretically grounded; computationally costly
Transparent design (Barredo Arrieta et al. 2020)	Inherently interpretable logic	No post-hoc approximation; constrains model choice

3.3 Financial NLP and News Representation

Measuring the similarity between news items requires turning text into vectors. Early work learned static word embeddings efficiently with word2vec (Mikolov et al. 2013), but these assign one vector per word, ignoring context. Contextual models based on the Transformer, notably BERT (Devlin et al. 2019), substantially improved language understanding, yet using BERT directly for semantic similarity is expensive because it compares sentence pairs jointly. Sentence-BERT (Reimers and Gurevych 2019) addresses this by producing sentence-level embeddings that can be compared efficiently with cosine similarity — the property this work relies on for precedent retrieval. For the financial domain specifically, FinBERT models fine-tune BERT on financial text, for sentiment (Araci 2019) and for broader financial communications (Yang, Uy, and Huang 2020). Table 3.3 compares these representations.

Table 3.3: Text-representation models for financial news.

Model	Representation	Strengths / limitations
word2vec (Mikolov et al. 2013)	Static word vectors	Efficient; no context, word-level only
BERT (Devlin et al. 2019)	Contextual token embeddings	Strong understanding; pairwise similarity is costly
Sentence-BERT (Reimers and Gurevych 2019)	Sentence embeddings (siamese)	Efficient cosine similarity; benefits from fine-tuning data
FinBERT (Araci 2019; Yang, Uy, and Huang 2020)	Financial-domain language model	Domain-adapted; oriented to sentiment / communications

3.4 Event Studies and News–Market Correlation

Quantifying the market impact of an event is the province of the *event study*. The methodology of Brown and Warner (1985) established how to measure abnormal returns around an event using daily stock returns, and remains the standard reference for the post-event return windows (e.g. +1, +3 days) used here. Building such an analysis historically requires news aligned to prices; the FNSPID dataset (Dong, Fan, and Peng 2024) provides exactly this — financial news time-aligned with prices for thousands of companies — enabling a news-to-impact knowledge base without scraping. The limitations are the usual ones of any dataset: coverage and time period, discussed in the data card.

3.5 Comparative Analysis and Positioning

Across these areas, the literature offers strong individual components but few integrated, explainable, retail-facing systems that combine them. The choices made in this work, and the alternatives they were weighed against, are summarised in Table 3.4. The guiding principle is *defensible simplicity*: where two options perform comparably, the more transparent and easier-to-explain one is preferred.

Table 3.4: Component choices in this work versus alternatives.

Component	Chosen	Alternative	Why
Anomaly detection	z-score (rolling)	Isolation Forest	Transparency / explainability
News retrieval	Sentence-BERT + cosine	word2vec / raw BERT	Efficient semantic similarity
Impact measurement	Event-study windows	—	Standard, reproducible
Explanation	Rules + precedents (+ SHAP)	LIME-only	Faithful, traceable

3.6 Summary and Research Gaps

The reviewed work supplies mature methods for each piece of the problem: transparent anomaly detection, efficient semantic similarity, domain language models, feature attribution, and event-study impact measurement. What is comparatively scarce is their *integration* into a single, explainable, reproducible system aimed at the retail investor — one that detects events and, for each, retrieves analogous historical news and presents their observed impact as traceable evidence. Addressing this gap, as an engineering contribution, is the purpose of the chapters that follow.

Chapter 4

Methodology

This chapter describes how the system is built. It presents the overall architecture and the two triggers, the two data layers, the method chosen for each component (with its justification), the evaluation design, and the rigour guarantees. Throughout, the guiding principle is *defensible simplicity*: where options perform comparably, the more transparent and explainable one is preferred.

4.1 System Overview and Architecture

The system is organised as a pipeline driven by two independent triggers: an abrupt market movement and an incoming news item. Figure 4.1 shows the components and the flow of data between them. The live layer supplies real-time prices and news; the historical knowledge base, built from FNSPID, supplies the precedents; the anomaly detector and the correlation engine produce the evidence; the explanation engine assembles a traceable account; and the Telegram bot delivers the alert.

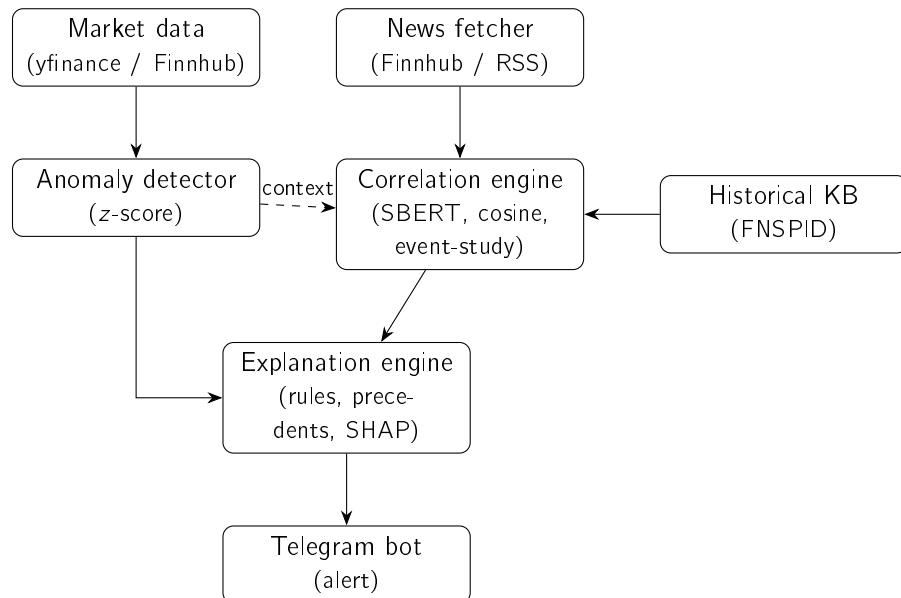


Figure 4.1: System architecture. Solid arrows show the data flow; the dashed arrow shows the anomaly providing context to the correlation engine. The two triggers (market movement via the anomaly detector; news via the correlation engine) converge on the explanation engine.

4.2 Data Architecture: Historical vs. Live

The system uses two clearly separated data layers. The **historical** layer is built offline from FNSPID (Dong, Fan, and Peng 2024), which provides financial news already time-aligned with prices; for each historical news item the system stores its text, date, ticker, a sentence embedding, and the observed post-event price impact. The **live** layer provides, in real time, prices (yfinance, with Finnhub as a fallback) and news (Finnhub and financial RSS feeds). Free news APIs are not used to build history because their free tiers are typically limited to recent windows and are delayed; the exact ticker subset, time window and preprocessing are documented in the data card.

4.3 Statistical Anomaly Detection

Abrupt movements are detected with a transparent statistical rule. Let r_t be the (log-)return of an asset on day t , and let μ_t and σ_t be the mean and standard deviation of returns over a trailing window of w days. The day is flagged as anomalous when the absolute z-score exceeds a threshold k :

$$z_t = \frac{r_t - \mu_t}{\sigma_t}, \quad \text{anomaly} \iff |z_t| > k. \quad (4.1)$$

This statistical family is standard (Chandola, Banerjee, and Kumar 2009) and is chosen for its transparency: the user can see exactly why an alert fired. More expressive detectors such as Isolation Forest (Liu, Ting, and Zhou 2008) were considered but not adopted as the default, because their reduced interpretability conflicts with the XAI-first goal; the window w and threshold k are documented design choices, evaluated in Chapter 6.

4.4 News–Market Correlation Engine

This is the core of the work. Each news item is encoded into a sentence embedding using a Sentence-BERT model (Reimers and Gurevych 2019), chosen because it allows efficient comparison of texts through cosine similarity. Given a new item, the engine retrieves the k most similar historical news items from the FNSPID-based knowledge base and reports the impact those precedents had. Impact is measured in the manner of an event study (Brown and Warner 1985): the asset’s return is computed over fixed post-event windows (e.g. +1, +3 days). The similarity metric and the impact windows are explicit design choices and form part of the contribution. Crucially, impact is measured strictly after the event, so no future information leaks into the features (Section 4.7).

4.5 Explanation Engine (XAI)

The explanation engine assembles a traceable account for every alert, combining three transparent ingredients: (i) the rule or measure that produced the detection (e.g. the z-score and threshold), (ii) the retrieved historical precedents and their observed impact as evidence, and (iii), optionally, feature attribution over the detector using SHAP (Lundberg and Lee 2017). This combination follows the explainability principles set out in the XAI literature (Adadi and Berrada 2018; Barredo Arrieta et al. 2020): the goal is an explanation the user can follow step by step. Optionally, financial sentiment from FinBERT (Araci 2019) may enrich the explanation; this component is cuttable if it does not add defensible value.

4.6 Evaluation Design

The evaluation is defined in advance, to avoid fishing for favourable results, and is reported in Chapter 6. In summary: the anomaly detector is assessed with precision, recall and F1 against an explicit definition of “true” anomalies, with a baseline and an ablation over window and threshold; the correlation engine is assessed on retrieval quality (precision@ k with a relevance rubric, against random and recency baselines) and on the measured event-study impact; and the explanation engine is assessed for fidelity to the actual logic and for usefulness, via a small human-judgement rubric. Because the system performs neither price prediction nor trading, profit-based metrics are explicitly out of scope.

4.7 Methodological Rigour

Three guarantees apply throughout. First, **no lookahead**: features at any instant use only information available at or before that instant, and historical impact uses only post-event windows; this is documented per experiment. Second, **reproducibility**: random seeds are fixed, dependencies are pinned, and every data/results figure is produced by a script in the repository. Third, **honesty**: only results that can be reproduced and explained are reported, and no data, metric or citation is fabricated.

Chapter 5

Implementation

This chapter details the implementation of the system: the environment and tooling, the historical data pipeline, the live layer, the core components, and the orchestration of the end-to-end flows.

5.1 Environment and Tooling

5.2 Historical Data Pipeline (FNSPID)

5.3 Live Layer: Market Data and News

5.4 Anomaly Detector

5.5 Correlation Engine

5.6 Explanation Engine

5.7 Telegram Bot and Orchestration

Chapter 6

Evaluation

This chapter reports the evaluation of each component against the design defined beforehand, together with end-to-end case studies, and discusses the results honestly, including their limitations.

6.1 Experimental Setup

6.2 Anomaly Detector

6.3 Correlation and Precedents Engine

6.4 Explanation Quality

6.5 End-to-End Case Studies

6.6 Discussion

Chapter 7

Conclusion

This chapter summarises the work, revisits the contributions, acknowledges the limitations, and outlines directions for future work.

7.1 Main Conclusions

7.2 Contributions Revisited

7.3 Limitations

7.4 Future Work

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Appendix A

Appendix Title Here

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